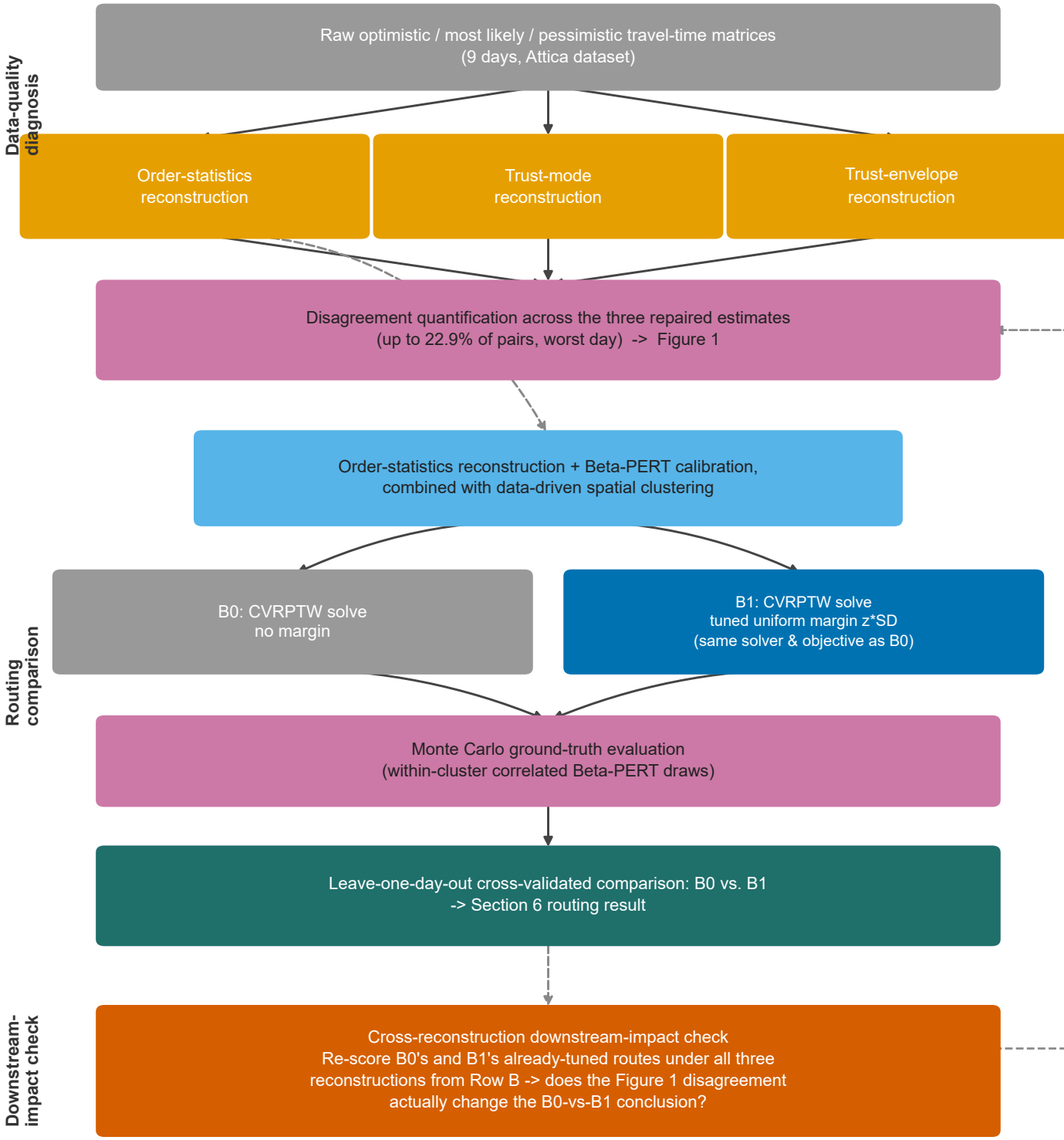Pipeline: reconstruction-ambiguity diagnosis and its downstream routing impact